# Data Science

## Lecture 07: Data Exploration and Preprocessing

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# Where We Left Off

Three lectures fitted models to tables nobody questioned. Today steps back to KDD steps 1–2 from Lecture 01: what the columns are, whether they can be trusted, and what a plot shows that a summary hides.

**Lecture 01**
data is never raw

→

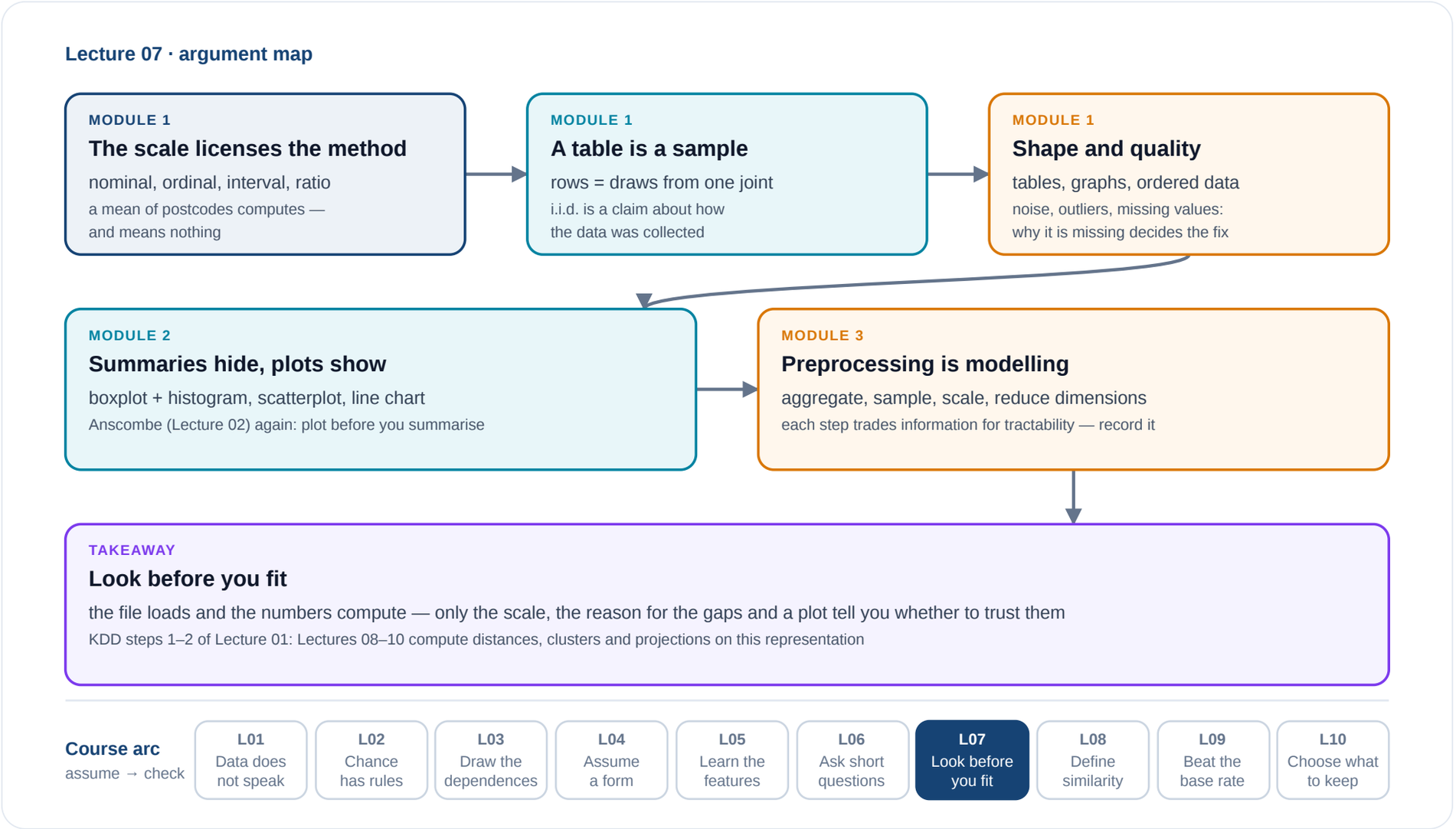
**Lecture 02**
summaries can agree while the data differs (Anscombe)

→

**Today**
scale, quality, plots, preprocessing

> From Lecture 08 on there is no label to check a model against — looking first stops being good practice and becomes the main check we have.

# Argument Map



# Module 1

## What Kind of Data Is This?

> ***Guiding Question:*** *The file loads and the numbers compute. Why is that not yet a reason to trust the average?*

# Four Attribute Types — What the Values Allow

| | type | the values… | meaningful | statistics | examples |
|---|---|---|---|---|---|
| categorical | **nominal** | are names that only distinguish objects | $=, \neq$ | mode, entropy, contingency table, $\chi^2$ test | ZIP code, employee ID, eye colour |
| | **ordinal** | carry an order | $<, >$ | median, percentiles, rank correlation, sign test | grades, rankings, {good, better, best} |
| numeric | **interval** | have meaningful differences; the zero is arbitrary | $+, -$ | mean, standard deviation, Pearson correlation, $t$-test | temperature in °C or °F, calendar dates |
| | **ratio** | have a true zero, so ratios are meaningful | $\times, \div$ | geometric mean, harmonic mean, coefficient of variation | length, mass, counts, age, temperature in K |

Each type permits everything the one above it does, plus more.

# Four Attribute Types — What You May Transform

| | type | permissible transformation | example |
|---|---|---|---|
| categorical | **nominal** | any one-to-one relabelling | reassigning employee IDs changes nothing |
| | **ordinal** | any strictly increasing $f$: $x \mapsto f(x)$ | {good, better, best} is as well coded by {1, 2, 3} as by {1, 5, 9} |
| numeric | **interval** | $x \mapsto ax + b$, $a > 0$ | °C and °F differ in zero point and unit |
| | **ratio** | $x \mapsto ax$, $a > 0$ | length in metres or in feet |

**INTUITION**

A statement is **meaningful** on a scale if no permissible transformation can change its truth. "Twice as long" survives metres → feet; "twice as warm" does not survive °C → °F.

# Why the Type Decides the Method

- ✗ **The classic error:** averaging a nominal attribute. The mean of ZIP codes 10115 and 81667 is a number, and it is meaningless.
- ✗ **The subtle one:** 20 °C is not "twice as warm" as 10 °C — interval scales have no true zero, so ratios do not transfer.

> The attribute type determines which **statistics**, which **distance measures** and which **plots** are permissible. Every method in this course depends on it.

# A Table Is a Sample From a Joint Distribution

| | payment | satisfaction | born | income (k€) | returned? |
|---|---|---|---|---|---|
| **column** | $X_1$ · nominal | $X_2$ · ordinal | $X_3$ · interval | $X_4$ · ratio | $X_5$ · nominal |
| **draw** $\mathbf{x}_1$ | card | 4 | 1987 | 52 | yes |
| **draw** $\mathbf{x}_2$ | invoice | 2 | 1995 | 31 | no |

Each **column** is a random variable **with a measurement scale**; each **row** is one draw $\mathbf{x}_i$ from **one joint distribution** $p(x_1, \ldots, x_5)$ that nobody has written down.

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| length recorded as | A | B | what the values support |
|---|---|---|---|
| cm — **ratio** | 180 | 90 | "twice as long" — in inches 70.9 vs 35.4, still ×2 |
| cm above 100 cm — **interval** | 80 | −10 | "90 cm longer"; the ratio −8 means nothing |
| size class 1–3 — **ordinal** | 3 | 1 | "longer"; "three times" means nothing |
| size label — **nominal** | L | S | "different" — nothing more |

# The Scale Suggests the Model

Which distribution would you write down for each column — and how many numbers does it need?

> The scale limits the model as it limits the statistics: a normal distribution for ZIP codes is as meaningless as their mean.

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| attribute | values | scale | a first model | numbers |
|---|---|---|---|---|
| returned? | yes / no | nominal | **Bernoulli**$(p)$ | 1 |
| payment | card, invoice, … ($k$) | nominal | **categorical**$(p_1, \ldots, p_k)$ | $k - 1$ |
| satisfaction | 1 – 5 stars | ordinal | categorical (order unused) | 4 |
| orders last month | 0, 1, 2, … | ratio, count | **Poisson**$(\lambda)$ | 1 |
| height | around 170 cm | ratio | **normal**$(\mu, \sigma^2)$ | 2 |
| income | $> 0$, skewed | ratio | **log-normal**: normal on $\log$ income | 2 |
| share of orders returned | $[0, 1]$ | ratio | **Beta**$(\alpha, \beta)$ | 2 |

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# What "i.i.d." Claims

The rows $\mathbf{x}_1, \ldots, \mathbf{x}_n$ are **independent and identically distributed** (i.i.d.) if

| | | |
|---|---|---|
| **independent** | $p(\mathbf{x}_1, \ldots, \mathbf{x}_n) = \prod_i p_i(\mathbf{x}_i)$ | *one customer's row tells nothing about another's* |
| **identically distributed** | $p_i = p$ for every row $i$ | *every row comes from the **same** joint distribution* |

Which of these datasets break it: daily sales of one shop · two visits by the same customer · pixels of one image · a survey sent to a random sample?

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Only the random survey is plausibly i.i.d. Time series, repeated measurements and spatial data all violate it — the arrangement *is* information.

# Four Shapes of Data

### Record data

Objects × attributes — the familiar table. Rows are *treated as* independent.

### Graph data

Objects carry **links** to other objects: web pages, social networks, citations.

### Fields / grids

Objects sit in a **spatial** arrangement: images, sensor lattices, simulations.

### Spatio-temporal

Position **and** time: monthly land temperatures, traffic movement, trajectories.

> The shape decides which methods even apply. A table whose rows are treated as independent can be shuffled without loss; a grid or a time series cannot — the arrangement *is* information. Independence of rows is an assumption, not a property of the file.

# Where Does a Recorded Value Come From?

A sensor records 22.4 °C. Besides the temperature, what went into that number?

$$x = s + b + \varepsilon$$

$s$: the true value

$b$: artefact — deterministic, the same every time

$\varepsilon$: noise — random, mean 0

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> A dataset records $x$, never $s$. Every data problem is a claim about the terms that were added on the way.

# Three Problems on One Axis




Which of the three problems does collecting more data fix?

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Only **noise**. An artefact shifts every new reading too; an outlier comes back in the same proportion.

# Each Problem Is a Model Statement

| problem | in the model | more data | what helps |
|---|---|---|---|
| **noise** | $\varepsilon \sim p(\varepsilon)$, $E[\varepsilon] = 0$ | averages it out | model it; average |
| **artefact** | $x = g(s)$, deterministic | does **not** help | calibration, a second instrument, domain knowledge |
| **outlier** | a draw from another $q$ | a share $\pi$ remains | robust statistics, or model $q$: $(1-\pi)\,p + \pi\,q$ |
| **missing** | $M$ = missing, $P(M \mid x)$ | depends on why | ask for the mechanism first |

None of these can be read off the data alone. Calling a value noise, an artefact or an outlier is an assumption about how it was generated.

# Mini-Exercise: Stars and Birth Years

A customer table records satisfaction (one to five stars) and year of birth. Which scale is each — and whose **mean** would you not report?

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Satisfaction is **ordinal**: its mean assumes equal gaps between the stars — report the median. Year of birth is **interval**: the mean year is fine, but ratios are not ("born twice as late" means nothing); differences such as age are fine.

# Mini-Exercise: Three Scales

Patients are weighed on three scales, a third on each. Two record $X = S + \varepsilon$; the third is miscalibrated and adds exactly 3 kg every time: $X = S + 3 + \varepsilon$. Does weighing ten times as many patients bring the mean of $X$ closer to $E[S]$?

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> No. $E[X] = E[S] + \frac{1}{3} \cdot 3 = E[S] + 1$ kg: the mean converges, but to a value 1 kg too high — an **artefact**. More data shrinks only the noise $\varepsilon$.

# Mini-Exercise: A Sensor That Fails in the Cold

Temperature $T$, missingness indicator $M$ (1 = reading missing). The sensor fails mostly in the cold: $P(M = 1 \mid T)$ is large for low $T$. After deleting the incomplete rows, does the mean estimate $E[T]$ — or what?

# Mini-Exercise: A Sensor That Fails in the Cold

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> It estimates $E[T \mid M = 0]$, which is **larger** than $E[T]$: the coldest days are gone. Only if $M$ is **independent** of $T$ is $E[T \mid M = 0] = E[T]$ and deletion harmless.

# Takeaway

> A dataset is $n$ **draws from an unknown joint distribution** — and every way of reading it is an assumption about how it was generated.

1. **Shape:** only record data can plausibly be read as independent rows.
2. **Table:** columns are random variables, rows are draws from $p(x_1, \ldots, x_d)$.
3. **Scale:** licenses a column's statistics and suggests its model.
4. **i.i.d.:** lets the rows factorise — to be checked, not presumed.
5. **Recorded value:** truth + artefact + noise. More data removes **noise only**; artefacts, outliers and missing values need a model of how they arose.

# The Churn Shop's Customer Table

| customer | contract | satisfaction | tenure (months) | postcode | churned? |
|---|---|---|---|---|---|
| 1 | monthly | 4 | 3 | 94032 | yes |
| 2 | yearly | — | 27 | 80331 | yes |
| 3 | yearly | 5 | 41 | 94032 | no |
| … | … | … | … | … | … |

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**SCALES**

Contract, postcode, churned: nominal. Satisfaction: ordinal. Tenure: ratio. The mean postcode computes — and means nothing.

**WHY IS SATISFACTION MISSING?**

Customers who left stop answering: 40 of 100 churners gave no rating, 45 of 900 stayers. Drop those rows, and churn falls from **10%** to **6.6%**.

# Module 2

## Seeing the Data

> ***Guiding Question:*** *Summary statistics answer the questions you thought to ask. What shows you the ones you did not?*

# Visualization Is Everywhere

Newspaper tables, subway maps, weather charts, stock graphs, a 3-D reconstruction of an injured knee, demographic bar charts, social-network layouts.

**What do these have in common?** All of them present data in a form the eye processes faster than the equivalent numbers.



# Why It Works

### Bandwidth

Vision is the human sense with the highest data rate — pattern detection runs in parallel and pre-attentively.

### Interaction

A picture is a natural surface for exploring: zoom, filter, select, compare.

### Discovery

Visualization surfaces the **unexpected** — the structure nobody thought to compute a statistic for.

Summary statistics answer questions you already asked. A plot lets the data ask you one — which is exactly what Anscombe's quartet demonstrated.

Visualization is not decoration for results; it is an **instrument of analysis**, and the only one that can show you what you did not think to measure.

# The Boxplot

A **visual summary of a distribution built from quartiles**:

### The box

Spans $Q_1$ to $Q_3$ — the interquartile range, the middle half of the data.

### The line

The **median** inside the box. Its position shows skew.

### The whiskers

(Tukey) Reach to the most extreme data points inside $[Q_1 - 1.5 \cdot \mathrm{IQR}, Q_3 + 1.5 \cdot \mathrm{IQR}]$. Points outside are drawn singly as **outliers**.

The 1.5 × IQR rule is a **convention**, not a test. It flags candidates for inspection, not errors for deletion.

# The Histogram

### What it shows

The **shape** of a distribution: modality, skew, gaps and clusters that a boxplot hides entirely.

### What it hides

Everything depends on the **bin width**. Too wide and structure vanishes; too narrow and noise looks like structure.

A bimodal distribution and a unimodal one can produce the same boxplot. Always look at one histogram before trusting a five-number summary.

# Mini-Exercise

Two datasets have the same minimum, $Q_1$, median, $Q_3$ and maximum — so the same boxplot. Must their histograms look alike?

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> No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

# Mini-Exercise

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No. The five numbers fix five points; the mass between them can be arranged freely — spread evenly inside the box, or piled into two clusters at its edges. Same quartiles, opposite stories: Anscombe's quartet again.

A boxplot compares distributions compactly; a histogram shows their shape. **Neither alone is sufficient**, and the boxplot's outlier rule is a convention you may overrule.

# The Scatterplot

An X–Y plot of two dimensions against each other — the standard instrument for inspecting a **relationship** rather than a distribution.

With more than two dimensions there are four options:

### Subsetting

Let the user choose which dimensions to display.

### Reduction

Project to fewer dimensions (see preprocessing, and projections later).

### Embedding

Map extra dimensions to other channels: colour, size, shape.

### Multiple displays

Superimpose, or juxtapose in a **scatterplot matrix**.

# What a Scatterplot Reveals

Linear association, non-linear structure, clusters, outliers and heteroscedasticity — most of which no single correlation coefficient reports.

**Pearson correlation** $r = \dfrac{\operatorname{cov}(X,Y)}{s_X s_Y} \in [-1,1]$: the covariance measured in units of the two standard deviations. It measures **linear** association only.

A correlation of 0 is consistent with a perfect parabola. The scatterplot shows it; the coefficient does not.

# Tufte's Principle

**Maximise the data-ink ratio:** every drop of ink on the page should carry information. Delete anything that does not — heavy gridlines, boxes, redundant tick marks, decorative fills. **Range frames** apply the principle to the axes: draw each axis only over the interval the data actually spans, so the frame itself reports the minimum and maximum.

> The exercise: take a finished chart and ask of each element, *what would I lose by deleting this?*

A scatterplot shows the structure a coefficient summarises away. Beyond two dimensions you must **subset, reduce, embed or juxtapose** — and every mark you draw should earn its ink.

# When a Line Is Justified

Connecting points with a line asserts an **order** between them — and often an interpolation.

> **Data requirement:** one or more quantitative value attributes on a comparable scale, plus one **ordered** key attribute (usually time).

> Connecting nominal categories with a line claims an order that does not exist. The line is a statement, not a decoration.

# Aspect Ratio Is Not Cosmetic

Readers compare trends by the **orientation** of the line segments, so the shape of the plotting area changes what they conclude. The same series looks flat in a wide frame and dramatic in a tall one.

**Banking to 45°** chooses the aspect ratio so that the segments' average absolute orientation is 45°: differences in orientation are judged best near ±45° (Cleveland et al. 1988).

### Additional channels

Colour and line type distinguish series — but only for a handful of them.

# Multiple Series

### Overlaid

Best for comparing a few series directly against each other.

### Stacked

Each series is drawn on top of the previous one. Shows the total well — and makes individual series **hard to read**, since each sits on a moving baseline.

### Sorted / small multiples

One panel per series, ordered by some property. Scales to many series where overlaying fails.

A line asserts order and interpolation; the **aspect ratio** decides what the reader sees and stacking trades the readability of the parts for the readability of the whole.

# Module 3

## Preprocessing

> ***Guiding Question:*** *Lecture 01 claimed that preprocessing is modelling. Which choices made here change what a model can learn?*

# Aggregation

Combine several objects or attributes into one — customer purchases summarised by store, by month, by price band.

### Gains

Fewer objects to process; **less variation**, hence more stable estimates; the natural operation of a data warehouse.

### Costs

Detail is gone, irreversibly. Effects that exist only within a group become invisible.

# Sampling

Work with a subset of the objects: random sampling with or without replacement, or **stratified** sampling that preserves group proportions.
Sampling always loses information. The question is whether it loses anything that matters — a sufficiently large sample retains the overall structure.

> Sample a sine wave once per period and every sample has the same value: the reconstruction is a flat line — not merely rougher, but **wrong**.

# Scaling and Encoding

## Scaling numeric attributes

Min–max: $x' = \dfrac{x - \min}{\max - \min} \in [0, 1]$

Standardisation: $z = \dfrac{x - \bar{x}}{s}$ — mean 0, standard deviation 1

## Encoding categorical attributes

**Nominal** → one-hot: one 0/1 column per value, so no false order.

**Ordinal** → ranks $1, 2, 3, \ldots$, which keep the order.

Two customers differ by 1,000 € in income and 30 years in age. Unscaled, their distance is $\approx 1{,}000$: age is invisible. Standardised ($s = 10{,}000$ € and $15$ years) the differences are $0.1$ and $2$ — now age dominates.

# Dimensionality Reduction

**The curse of dimensionality.** As the number of attributes grows, the data becomes sparse: distances between points concentrate, neighbourhoods lose meaning, and the volume to be covered grows exponentially.

### The problem

Many features are irrelevant, noisy or correlated.

### The cost

More computation, more parameters, more variance — and weaker distance-based methods.

### The response

**Feature selection** keeps a subset; **feature extraction** builds new combined dimensions.

Every preprocessing step **trades information for tractability**. The trade is usually worth it — but it is a modelling decision, and it should be recorded as one.

# Image Credits & References

Cleveland, William S. and McGill, Marylyn E. and McGill, Robert (1988). The Shape Parameter of a Two-Variable Graph. *Journal of the American Statistical Association*.